

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456068.861 +/- 0.052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.053
Transit depth (Rp/Rs)^2	2.01 +/- 1.5 [%]
Orbital Inclination (inc)	89.81 +/- 2.16
Airmass coefficient 1 (a1)	1.363 +/- 0.019
Airmass coefficient 2 (a2)	-0.1481 +/- 0.006
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	None
Optimal Aperture	4.28
Optimal Annulus	10.27
Transit Duration (day)	0.076 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.053 vs 0.1457 (deviation 0.07σ)
Duration fit vs published	0.076 d vs 0.12 d (deviation 1.26σ)
Mid-time O–C	-5.4 min
Depth SNR	2.7
Residual scatter	1.33 %

Light curve

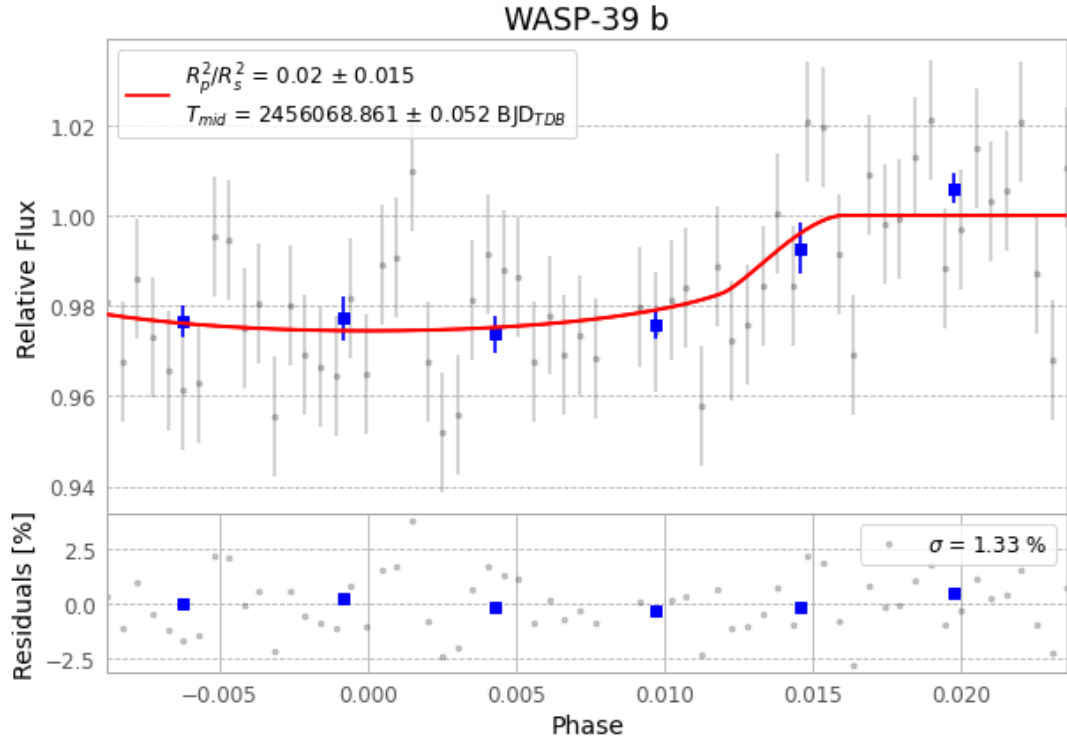


Figure 1 — FinalLightCurve_WASP-39 b_2012-05-21.png (SHA-256 457cbf3b5bf4283e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 190bc16468a64a90...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

62 FITS frames (41 MB), WASP-39120521073913.FITS → WASP-39120521104813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-120521.log (3d5655e2397c...), FinalLightCurve_WASP-39 b_2012-05-21.png (457cbf3b5bf4...), FinalLightCurve_WASP-39 b_2012-05-21.csv (5e1c70d71caa...)

Compute resources

Wall time 82.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:18Z.